KATHIRESAN MURUGESAN

Software Engineer | React.js · JavaScript/TypeScript · Node.js · PHP · MySQL · AI/LLM

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building and deploying production web applications using React.js, JavaScript/TypeScript, Node.js/Express.js, PHP, MySQL and PostgreSQL. Delivered 8+ full-lifecycle platforms across EdTech, food-security analytics, AI document processing, ERP and fleet management, improving user engagement by 25%. Hands-on with AngularJS, OpenAI and Claude API integration, Stripe/PayPal payments, role-based access and WebSocket real-time features. Works closely with AI/Python engineers, product managers and QA teams.

TECHNICAL SKILLS

Frontend: React.js (Hooks, Context API, React Router), AngularJS, JavaScript (ES6+), TypeScript, HTML5, CSS3, Material UI, Bootstrap, Recharts, jQuery/Ajax

Backend: Node.js, Express.js, RESTful APIs, JWT, WebSocket, PHP (Core PHP, Laravel)

AI & Integrations: OpenAI API, Claude API, Prompt Engineering, Stripe API, PayPal API, Xero/QuickBooks API Webhooks

Database & Cloud: MySQL, PostgreSQL, Schema Design & Tuning, AWS S3

Tools & Practices: Git/GitHub, Postman, VS Code, Agile/Scrum, Unit Testing, Debugging, Code Reviews

PROFESSIONAL EXPERIENCE

Software Engineer (Full Stack Developer) | May 2022 - Present

Cap Digisoft Solutions Pvt. Ltd., Coimbatore, India

- **Delivered 8 full-lifecycle platforms** across EdTech, food-security analytics, AI document processing, fleet inspection and order management, owning requirements through deployment.
- **Improved user engagement by 25%** by building responsive, high-performance React.js applications backed by Node.js/Express.js and PHP APIs.
- **Integrated OpenAI and Claude APIs** and applied prompt engineering to add AI-assisted features, working closely with AI/Python engineers.
- **Designed and shipped RESTful APIs** with Node.js/Express.js and PHP; integrated Stripe and PayPal for secure payments, including bulk payments and webhooks.
- **Modernized an 8-year-old legacy codebase** by migrating the frontend from React 15 to React 19 and resolving deprecated packages and dependencies.
- **Developed analytics dashboards** with Recharts and standard-deviation indicators for UN FAO food-security reporting across 6 countries.
- **Optimized MySQL and PostgreSQL schemas** for integrity, security and query performance in multi-tenant applications; built AWS S3 file and media pipelines.
- **Reduced post-release defects** through systematic testing, debugging and code reviews; shipped on time with project managers, instructional designers and QA.

KEY PROJECTS

Attesta AI - AI Document Processing Platform | ***React.js, Python AI backend, WebSocket***

Automated summaries, OCR and requirement-based categorization of scanned documents. Role: complete React.js frontend.

- **Upload & tracking:** drag-and-drop upload with validation and progress; live AI status via polling and WebSocket.
- **Results & configuration:** dashboard for AI summaries and extracted fields; panels to define extraction rules, field mappings and categories.

UN FAO Food Security Monitoring System | *React.js, Node.js/Express.js, PostgreSQL, Recharts*
Dashboard visualizing food-security indicators across 6 MASHREQ-region countries.

- **Frontend:** reusable Recharts components for multi-country comparison with colour-coded above/below standard-deviation indicators; Context API and custom Hooks.
- **Backend:** Node.js/Express.js APIs filterable by country, indicator and year range; PostgreSQL schemas; server-side statistics.

MyGooRoo - Tutor App | *React.js, Node.js/Express.js, MySQL*
EdTech platform for tutors to manage diagnostics, curriculum, teaching tools and online classes.

- **Frontend:** reusable React components for tutor dashboard, session management, quiz builder and PDF annotation; role-based UI with protected routes.
- **Video & backend:** video conferencing with attendance auto-synced on session start/end; Node.js APIs and MySQL schemas.

MyGooRoo - Student App | *React.js, Node.js/Express.js, MySQL*
Gamified, curriculum-aligned learning platform with personalised paths and live progress tracking.

- **Gamification:** coin rewards, avatars, leaderboards, streaks and subject-wise progress dashboard.
- **Learning tools:** worksheet annotation (highlight, draw, text) with homework submission and spaced-repetition flows.

MyGooRoo - Tutoring Centre Management System | *React.js, PHP, MySQL, Stripe API*
Role-based platform for admin and centre users covering scheduling, fees and communication.

- **Frontend:** drag-and-drop event calendar, dynamic forms and PDF reports with protected routes.
- **Backend:** PHP/MySQL REST APIs; Stripe payments for fees and discounts; modules for students, parents, tutors, leads, sessions and notifications.

AVP - Order Management System | *React.js, PHP, MySQL, PayPal API*
ERP-integrated system for commercial and non-commercial orders with admin controls.

- **Build:** React.js order entry, customer management and live status tracking; PHP backend for estimation, invoicing, advance and bulk payments.
- **Payments & reports:** PayPal REST API integration, bill receipts, item tracking and reporting.

iLedger - Vehicle Inspection Management System | *PHP, MySQL*
Role-based fleet inspection system for vendors, drivers and mechanics across 9 truck types.

- **Workflow:** Pre-Trip, Post-Trip, Mechanic Response and Completion with role-based dashboards; visual truck-parts display with Good / Medium / Critical status and action items.

LINCtrac Platform (LTP) - LINC Education | *React.js (React 15 to 19 migration)*
Migrated an 8-year-old legacy frontend from React 15 to React 19 and resolved deprecated packages and dependencies.

Practica - School & College Management App | *PHP, jQuery/Ajax, MySQL, JSON*
Chat-based platform bridging offline and online learning for teachers and students on any device.

- **Modules:** attendance, assignments, notices, timetable and progress reports with JSON-driven data exchange.

EDUCATION & LANGUAGES

Master of Computer Science | 2020 - 2022 · 81%
Bharathiar University, Coimbatore
PG Project: Mobile App for Effective Water Management System (ESP8266 NodeMCU, Blynk cloud platform)

Bachelor of Computer Science | 2017 - 2020 · 70%
Sri Vasavi College, Erode
UG Project: Printing Office Automation System (VB.NET, MySQL)

Higher Secondary (HSC) | 2017 · 75.42%
Govt Boys Higher Secondary School, Thiruchengode, Namakkal

Secondary School Leaving Certificate (SSLC) | 2015 · 84.8%
Sankar Higher Secondary School, ICL, Namakkal

Languages: Tamil (Native) · English (Professional Working Proficiency)